



Module 14

Construction of a Training Program

A training program must be flexible and dynamic. Without ongoing assessments and modifications, a training program quickly can be deemed ineffective. The quality of a coach is not so much judged by the ability to create a training program, but rather by the ability to modify and manage it to get the most out of a client. This requires weekly feedback from clients to ensure programs are relevant and on-track in relation to the goals.

Clients will come to you with differing abilities, experience levels, goals, and starting points. Perhaps your client was working solo or with other coaches but not getting the desired results and is now three months away from the goal race. Conversely, your client may be new to exercising, unable to run more than one mile and completely deconditioned. The problem with many training programs is that they make assumptions. Some of the more common client assumptions are:

- Form and biomechanics are correct
- Base level of proficiency in regard to skill and technique
- Training program starts at the beginning of a calendar year
- Training at a particular volume prior to the start of the program
- Prior running experience
- In good physical condition
- Training for a particular event

Most training programs focus on training volume. This is primarily because if a client is unable to complete distances in training that are at or near those of the event being trained for, the likelihood of successful completion on race day is slim.

This emphasis on volume versus intensity is typically greater with new and/or deconditioned clients.

Training programs range from generic to extremely involved and detailed. The type of training programs you create for your clients are largely based on your clients' goals and should also consider their strengths and weaknesses.

An important thing to remember when developing a training program for a client is that your client is not you! Many coaches design programs around what they like and what has personally worked for them. While this approach may work in some cases, it is important not to construct training programs this way. Individuals will have varying responses to the same training stimuli. This is a critical piece that separates a qualified coach from an unqualified coach. A qualified coach will construct a training program that is grounded in science and customized for an individual client.

LACK OF RESEARCH

Despite the many articles in magazines and online regarding training programs for runners, there are a surprisingly low number of peer-reviewed research articles on this subject.

Therefore, most articles regarding preformatted programs and training program construction are either based on the foundations of exercise science and/or based on the past experience of coaches regarding their personal experience and those they have coached.

This is not necessarily a bad thing. Oftentimes the training practices that coaches implement are done because they know they work based on past experience, even if they don't know the physiological rationale behind the training methodologies. In other words, the role of sports science is often to explain why a particular training methodology works versus developing a new and groundbreaking training theory.

Because of this, the majority of the content in this module is formulated via applied exercise science and evidence-based findings from our team of advisers, versus solely running programming studies.

❖ OBJECTIVES

Upon completion of this module, you should have an understanding of the following areas:

- Why preformatted training programs are not recommended
- Elements of running recovery
- Components of a training program
- Steps to create a training program
- Maximum recommended weekly training hours
- Challenge to the 10 percent volume increase rule
- Maximum weekly increase in volume guidelines
- Guidelines for maximum individual session volume increases
- Weekly sport-specific distribution
- When over-distance training is appropriate

The most important thing to realize when creating a training program is that there is no set formula. With so many variables, coaches must default to their knowledge of the body and their clients to create safe, progressive, and effective programs.

To illustrate this point, if you have 20 clients all training for the same race and starting training at the same time, all of the programs should be different.

Many coaches envision program creation as plug-and-play. While it is possible to use templates to create programs, this process almost all but guarantees that a program will fall short in some capacity as it is not customized.

Therefore, don't expect to go off set numbers in regard to volume increases and decreases or predesignated weekly workouts. To provide real value to your clients, you must customize the programs in all areas.

Preformatted Training Plans

Pick up any running magazine or training manual and you will no doubt find that it contains training programs. They give specific plans to follow to achieve one's desired results. While one of these plans might work well for one person, it probably does not work so well for others. In other words, preformatted training plans are akin to a "one size fits all" model.

The primary trait that most preformatted programs have is a predetermined starting training volume. For example, a marathon-training program might assume that an athlete has been running 20 miles per week for the previous four weeks. This may or may not be the case. In regard to predetermined starting training volumes, this approach works only with someone not being professionally coached. Using the preceding example, an individual without a coach would train until reaching 20 miles/week and then would begin following the program. However, in a coach/client relationship, this buildup period to 20 miles/week is part of the training program. Therefore, the actual training program would be longer than noted in the preformatted plan.

This certification will not have preformatted training plans as part of the content. Why? Preformatted programs simply do not adjust for a client's specific needs. Following are several variables that cannot be accounted for with preformatted training plans:

- Recovery rate
- Accurate training volume
- Sport-specific strengths/weaknesses
- Time availability
- Physiological adaptation to training (runners adapt to the same training program at different rates and in different ways)
- Injury

It is incorrect to force potentially hundreds of different training scenarios into just a few preformatted training programs. This leads to programs that are not individualized and, therefore, nowhere near as effective as they could be. Training programs need to be individually designed for each client – period. Preformatted training programs are a great place to get ideas but should not be used "as is" for clients unless it happens to fit exactly with what a client needs.

❖ DELIVERY OF PROGRAM

Training programs contain a lot of information. There is often so much information presented to a client all at once that it leads to sensory overload and a lack of comprehension. Conversely, if a coach oversimplifies it, the program often leaves out critical details. What UESCA recommends is a training program that comprises three components:

1. **Full “At a Glance” Program**
2. **Weekly Training Programs**
3. **Workout Database**

By breaking down a training program into these three parts, a client is able to view the program on multiple levels and is better able to understand the individual elements of the program and how they interrelate.

A training program works only if it is adhered to. Clients who do not stick to the plan by skipping workouts or by doing their own thing risk not meeting their goals. This is where a training log that is reported to the coach is of paramount importance, as it places accountability on the client.

As noted previously, most preformatted training programs have some sort of prerequisite in regard to existing mileage before starting a program. The exceptions are programs that assume the individual has no running experience and therefore is not currently running. A popular program such as this is the *Couch to 5K*®.

Following are the standard durations of preformatted programs (in weeks) that require some sort of mileage prerequisite:

- **Marathon:** 16–18 weeks (often relates to an 18-to-20-miles/week prerequisite)
- **Half Marathon:** 10–12 weeks (often relates to a 12-to-15-miles/week prerequisite)

While some preformatted programs are longer or shorter, it is important to know that these training durations are often what a client may have in mind. If a client is new to running and/or deconditioned, the training program duration will likely

be much longer as the person does not have an existing mileage and fitness base.

❖ PROGRAM DESIGN TRENDS

Following are general trends concerning running program design:

- 1. From the inception of training to the goal event, weekly training volumes gradually increase until the taper begins.**
 - a. Experienced runners will have more long runs in their program than intermediate and novice runners.
 - b. The longest run is typically integrated into the week before the taper.
 - c. Beginner runners typically perform their first long runs later in the program than intermediate and experienced runners.
- 2. Deconditioned clients generally start at a low volume and intensity level and spend the majority of the training program focused on building fitness and endurance.**
 - a. Clients who begin a program at a moderate to high level of fitness typically have a reduced focus on fitness and aerobic development as compared with a deconditioned client.
 - b. Clients with a relatively high level of fitness at the inception of a training program typically integrate high-intensity-level training sooner than those with lower beginning-fitness levels.
- 3. High-volume weeks are typically followed by a recovery week, especially with novice runners.**
- 4. As clients progress through the training process, they should be able to handle a higher workload with faster recovery.**
- 5. Generally speaking, there should not be more than two consecutive weeks of increasing volume unless the training volume is low.**

❖ RECOVERY

While training programs can be quite complex, the genesis of a training program is to stress the body, allow it to recover, and then stress it again. This process, repeated over and over, will allow an individual to improve health and athletic performance. The goal is to arrive at race day with the best possible fitness level, a high level of technical proficiency, and be well rested!

EASY/HARD DAYS

Recovery between workouts is important for many reasons. Some of these reasons are:

- Decrease the chance for injury/illness
- Get the most out of future workouts
- Reduce the chance for overtraining

While a client's ability to recover must be assessed on an individual basis, it is safe to say that after any hard training day, a recovery day should follow. A recovery day may constitute an easy run or a day off. "Hard days" traditionally constitute any form of training that puts a lot of stress on the body (ex: long runs, intervals, hill repeats).

"Hard days" are subjective. For a professional marathon runner, running 17 miles at a subthreshold pace may not constitute a "hard day," while for most everyone else who does not run for a living, it most certainly is!

Because of the individualized rate of recovery, it is difficult – if not impossible – to decipher what the optimal amount of recovery (in days) is after a hard workout. There are so many variables that the best way to determine this is based on the history and progression of a client.

One day's rest after a hard day might not be sufficient for an individual to be totally recovered from a physiological standpoint. Think about it. A marathon taper is typically three weeks in duration. Why? The theory is that it takes this long for the body to recover from all the training and to build up proper glycogen stores. It is highly probable that physiologically, the body is not completely recovered one or even several days after a hard workout. Also keep in mind that

fatigue is cumulative. As a result, it is not just a singular hard workout that the body is recovering from but rather the total workload of training over a period of time.

Therefore, unless your client is tapering for a race, the goal is not to be totally recovered physiologically but rather recovered enough to be able to perform a prescribed workout at the desired pace/distance without risking injury or illness.

Back-to-Back Hard Days

As noted previously, athletes will adapt to the same training stimulus in different ways. Some athletes have found that performing back-to-back hard workout sessions (high volume or high intensity) followed by successive recovery days helps to elicit a greater training adaptation response than training one hard day followed by one easy day. The exact number of recovery days that follow are based on the individual's recovery rate. However, if three or more recovery days are required, back-to-back hard days should not be implemented in a program.

This type of training should be attempted only by an athlete who recovers well on a day-to-day basis.

SLEEP

Many physiological changes take place during sleep, such as glucose and blood pressure changes. There are also hormonal changes that according to Dattilo et al. may play a large role in a muscle's ability to recover and get stronger (721). Sleep deprivation leads to an increase in cortisol and a reduction in testosterone. As a result, protein synthesis, which is required for rebuilding damaged muscles, is greatly reduced (721).

Another study by Res et al. found that ingesting protein right before sleep aids in post-exercise recovery by stimulating muscle protein synthesis (671).

DO WE REALLY KNOW WHAT TRAINING VOLUME IS OPTIMAL?

To some degree, there is a trickle-down effect from professional runners to age-groupers in regard to training theory and methodology. As previously mentioned, this is largely due to magazine and online articles about how pros train. Of course most age-groupers do not have the time to train at the same volume as the pros, and even if they did, most could not handle the volume without getting injured. However, the message is clear in regard to running marathons: If you want to be the best, you need to run a lot of miles ... period. Most elite marathon runners average between 100 and 125 miles per week (717, 719).

Is more always better?

Likely not. When interviewed, many elite marathon runners often state that they are extremely conscious of the importance of recovery and therefore are careful not to overtrain. It seems as if they have identified a “sweet spot” that allows them to put in enough miles to elicit the desired training adaptations but not at the risk of becoming overtrained. So the question is, even at this self-selected “sweet spot,” could they be running fewer miles and still perform at the same level ... or even faster? The following diagram illustrates that the ideal training volume and intensity exists on bell curve, meaning too much or too little in regard to volume and intensity will likely result in decreased performance. The correct volume and intensity must be assessed and determined on an individual basis.

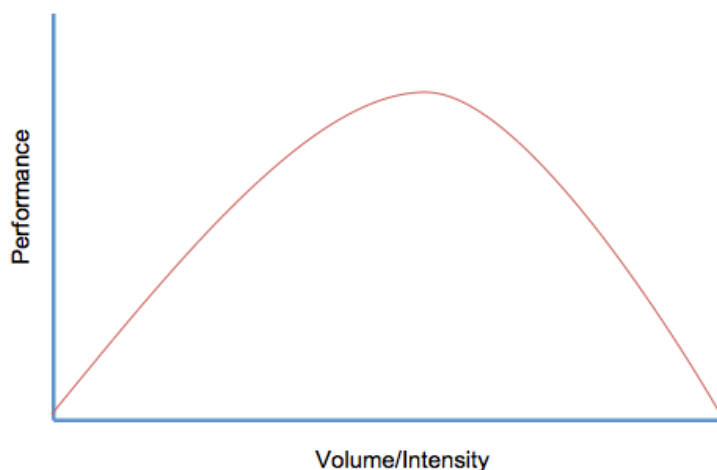


Figure 14.1 Volume/Intensity Bell Curve

NOT ALL MILEAGE HAS THE SAME VALUE

According to running coach Jack Daniels, increases in training volume past a certain point follow the law of diminishing returns (720). This means that increasing volume to a certain point will substantially benefit a runner. However, volume increases past this point will have a reduced benefit. In other words, you don't get the same bang for your buck. This is a slippery slope as not only are the benefits reduced, but the chance for injury also increases. This relates to the "sweet spot" and the associated diagram (figure 14.1).

CONFLICTING INFORMATION

Historically, elite marathon runners have always trained at a high volume and therefore it has been established that this is a prerequisite to running a fast marathon. At the elite level, the phrase *If it ain't broke, don't fix it* is very applicable. For an age-grouper, changing a training program or strategy has very few repercussions. However, for professionals, there is likely too much risk in drastically altering an annual training regimen. This is likely why there is not a lot of research in the area of varied training volume among elite marathoners. As a result, high weekly mileage remains a constant among elite marathoners.

Several studies suggest that only high-intensity training leads to increases in fitness among well-conditioned aerobic athletes, and that low-intensity training does not lead to increased aerobic capacity. As the majority of time spent running by elite marathon runners is at subthreshold levels, their training methodology seems to contradict the science behind improving one's cardiovascular capacity.

Conversely, there is evidence that an increase in running volume correlates with an increase in running economy (efficiency). Running long miles also increases one's muscular endurance and mental stamina. So is the mileage run by elite marathoners a necessity or largely based on historical patterns of training, and how does this information relate to the average runner? At this point, no one really knows the answer.

Avoid Injury

In order for an individual to finish a race, it requires being able to start it. If your client is injured and unable to start a goal race, it doesn't matter how great the previous training was. Not all injury is due to overtraining, and more specifically,

there is a certain degree of randomness to training and injury. Above all else, it should be your goal to get your client to the start line on race day!

LONG RUNS

At the most simplistic level, more time spent running equates to more stress on the body. This is especially the case with long runs.

Individuals who introduce high-mileage days before their body is adapted put excess stress on their musculoskeletal system and therefore are at an increased risk for injury. This is especially the case for runners who have biomechanical inefficiencies and imbalances. These issues likely increase the chance for injury over those who are biomechanically sound.

A “fast” runner may be able to clip off 20 miles in training at a seven-minute-mile pace (2:20 elapsed time) while another runner may take 3:20 (10-minute-mile pace) to cover the same distance. The stress on the body for the slower runner is much greater than that of the faster runner, primarily because of duration of time spent running (one hour more).

In regard to marathon training, the average prescription for the maximum-distance long run is 20 miles. Therefore, for a slower runner, it may be advised for the long run to be 18 miles. Conversely, for faster runners and those who have a low incidence of injury, 22 miles may be the optimal long-run distance.

Conclusions

A well-constructed training program must sufficiently stress the body but also allow for proper recovery. In regard to intensity versus volume, the ratio between the two must be configured on an individual basis. While long training runs are an important aspect to a marathon program, it is likely that an athlete in good to great aerobic condition could benefit from increased intensity as much or more from increased volume.

As a general trend, volume and intensity have an inverse relationship (figure 14.2). Therefore during a week of volume increases, it is advised not to schedule large amounts of speedwork.

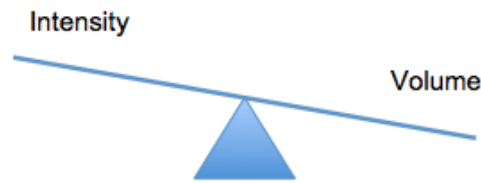


Figure 14.2 Volume/Intensity Balance

❖ OTHER PROGRAMMING FACTORS

TIME vs. DISTANCE (Long-Run Specific)

Many factors influence one's ability to tolerate training stress on the musculoskeletal system. You probably know a runner who can barely even look at a pair of running shoes without getting injured while another runner can seemingly run endless miles without ever getting so much as a blister! These two examples should be considered outliers. The rest of us fall somewhere in between.

Regarding long runs, some advise that the duration be prescribed by time versus distance. Famed running coach Dr. Jack Daniels advises that for recreational runners (nonelite), the long run should be no longer than 2:30. His theory is that because of the lack of adaptation on the body from a musculoskeletal point of view, running over 2:30 would increase the chance for injury to the point where it's not worth the gamble (757). Moreover, Dr. Daniels states that even if an individual's goal time is 4:30 and the person's longest run is 2:30, this volume disparity should not be cause for alarm, as it is not a necessity to train at or close to the race distance (757). In other words, Dr. Daniels looks at the long run from the perspective of stress on the body. For example, if a top-level marathoner runs the marathon in 2:20 whereas a recreational runner does it in 4:40, from purely a time perspective, the recreational runner places twice the amount of stress on the body as the top-level marathoner.

If prescribing long-run volume by time, the exact maximum should be determined by many factors, including training history, comparison between prescribed long-run time, and estimated marathon finish time and propensity for injury.

ARGUMENT AGAINST USING TIME-BASED DISTANCES

While nothing is inherently incorrect about prescribing a time-based long run, if the prescription is substantially below what a runner expects to run during the race, this likely poses several issues.

Issue #1:

If a time-based long run is prescribed because you are worried that the client cannot handle the standard long-run distance without a substantial chance of getting injured, what does this say about the person's overall readiness for the marathon?

Remember, the first goal of the runner is to finish the marathon, irrespective of a target finishing time. If the time covered in training is drastically less (over half) than what is expected on race day, a client is venturing into unknown territory as the person doesn't know how his or her body (and mind) will respond to the increase in time/miles. It may respond fine ... but it may not.

Issue #2:

Running a marathon has as much to do with the mental aspect as it does the physical aspect. Let's say that a runner performs a long run of 2:30 but projects the marathon finish time to be five-plus hours. Even if the client could handle this increase in distance without physical injury, would the person be able to mentally handle the increased time on his or her feet? Likely not.

Summary:

Because of both the physical and mental factors involved with training for a marathon and specifically the long run, UESCA recommends a long-run prescription range of 18–22 miles. If a time-based long run is prescribed, it should roughly equate to the time required to run within the 18-to-22-mile range.

FACTORS THAT INFLUENCE VOLUME INCREASES

There is little argument that substantial increases in training volume (both long run and weekly volume) from week-to-week will likely increase the chance for

injury. What qualifies as a *substantial increase* is subjective, as it is based on the individual. Many factors influence volume increases within a training program. Following are the most influential:

- **New Runners:** Musculoskeletal system has not adapted to the impact of running.
- **Low Training Volume:** Body has not adapted to high volume.
 - Individuals at a high training volume typically tolerate large volume increases better than those at low training volumes.
- **Physiological Adaptation:** The speed at which your client adapts to cardiovascular and musculoskeletal training will affect how fast the individual can progress. Of the two, musculoskeletal adaptation is more significant in respect to volume progression.
- **Biomechanics:** If your client has substantial postural or biomechanical deviations that would likely increase the chance for injury, volume increases should be made sparingly or not at all and only after consultation with a specialist.
- **History:** If your client is an experienced runner, you should note the individual's long-run volume increases during previous years and the results (e.g., performance, injury, etc.) to help estimate the appropriate volume increases. Additionally, if there were any reasons for delays in the client's program progression such as injury or burnout, this should also be noted.
- **Injury:** If your client is recovering from an injury or sickness, initial volume increases, if any, should be minimal.
- **Current Training Volume:** Increases in volume are largely dependent on the current training volume of the client.
- **Genetics:** Some clients are naturally gifted in the area of endurance and musculoskeletal adaptation. However, until you work with a client for a substantial amount of time, you will not be able to determine this.
- **Ability to Recover:** Many factors affect one's ability to recover, including proper rest, nutrition, manageable volume, and intensity increases.

The date or distance of an event is not a factor in increasing volume. Volume should never be increased to accommodate an event. If based on current training volume and proper volume progression a client will not be ready for an event, a new event must be selected.

THE 10 PERCENT RULE (Fact or Fiction?)

A 10 percent maximum increase in volume is commonly recommended for running programs. Most coaches reference and use the 10 percent rule to steer their clients away from overtraining and potential injury. So where did this popular guideline originate from? No one knows. According to Carl Foster, director of the Human Performance Laboratory at the University of Wisconsin–La Crosse, the origin is “lost in history,” and he notes that “whether it (the 10 percent rule) is right is undocumented” (130).

Challenge to the 10 Percent Volume Guideline

The genesis of the 10 percent volume increase rule is to reduce the risk for injury. A 2008 study out of the University of Groningen in the Netherlands challenges this notion. There were 532 subjects in this study, split into two groups to train for a four-mile running race (130). One group of runners was put on a 13-week program with 10 percent weekly increases in volume. The other group was put on an eight-week program with 50 percent weekly increases in volume. The study found that the 10 percent increase group had a 20.8 percent injury rate, whereas the 50 percent increase group had a 20.3 percent injury rate. In other words, the two groups essentially had the same injury rate, approximately one in four.

While this study seems to indicate that the rate of volume increase is not a factor for injury, it is important to note that many factors contribute to injury, and thus, volume increases must be based on the individual.

A 2012 study by Nielsen et al. also turned up similar findings (316). In this study, 60 novice runners tracked their distance over 10 weeks using GPS technology. Of the 60 runners, 13 sustained an injury. Those injured increased their weekly training volume 31.6 percent in comparison with those who were not injured (22.1 percent). While this study seems to demonstrate that there is an increased chance for injury with large increases in weekly training volume, it needs to be noted that the uninjured individuals increased their weekly training volume 22.1 percent, which is much greater than the customary 10 percent.

Dr. Ron Diercks from the University of Groningen in the Netherlands, who was a member of the team that conducted the study mentioned above, noted that in regard to the 10 percent rule, “Nobody found out if it works or what is the basis of it.” Dr. Diercks goes on to say, “People hear something, they read something, and then it’s like a religion” (130).

Given the lack of an origin or scientific explanation of the 10 percent rule, it appears to be largely based in running and training lore.

When using a set percentage (10 percent) for the purpose of volume increases, the initial increases will be quite small (assuming the client is starting out at a low

training volume), and as the volume increases, the increases get larger and larger. There are two primary issues with this approach:

1. Increases in volume are based on a preset, arbitrary percentage, and therefore do not take into consideration an individual's adaptation response to training.
2. If a client were starting at a very low training volume, the initial increases would be so small that they would be deemed inconsequential.
 - a. For example, if your client is starting out running one mile, a 10 percent increase is .1 mile. Based on the 10 percent increase logic, the client would run 1.1 miles the following week.

Clients respond differently to the same volume increases.

While this section debates the validity of the 10 percent rule, it must be noted that weekly increases of less than 10 percent (or no weekly increases) are standard in training programs.

WEEKLY VOLUME INCREASES

Weekly volume increases are distributed unevenly across a week's worth of training. For example, if a client is running 40 miles per week, a 10 percent increase equates to a four-mile weekly increase for a total of 44 miles. Below is how the four miles might be distributed:

Long run: 1.5 miles (13 to 14.5 miles)

Tempo run: 1 mile (5 to 6 miles)

Easy run: 1 mile (4 to 5 miles)

800-meter repeats: .5 mile (one more repeat)

As the four additional miles are spread unevenly across four days (workouts), when you distribute the increase across X number of days, the daily increases are likely not substantial. While large weekly increases in volume may or may not increase the chance for injury, when discussing percentage volume increases, it would be advantageous to focus on daily volume increases, as this likely has a stronger correlation to injury rate than weekly volume increases.

Weekly volume increases can also be accommodated via an additional day(s) of running.

❖ PROGRAM CREATION OVERVIEW

On the surface, creating a training program is a fairly straightforward and intuitive process. However, with so many variables, it is often difficult to know exactly where to start and how to progress the program. As you never know who will want your coaching services, you cannot afford to make any assumptions (i.e., currently running 15 miles/week) when creating a training program. While it is fine to use some form of template for your clients, the template must be completely customizable to the needs of each client and make no assumptions.

Following is the step-by-step process by which UESCA recommends creating a training program. Each step is elaborated on later in the module.

Before you can take on clients, you *must* make sure that based on their desired goal(s), their current training level, and their availability to train, they will be well prepared for their event(s). If after taking these variables into account, you find that they will not be ready, you must either select a more reasonable goal or refuse to work with them.

Program creation comprises two areas:

1. Feasibility mock-up

- a. Takes into account the distance of the goal event, client's current fitness level/training volume, and weekly training availability

2. Full training program build-out (once the program is deemed feasible)

- a. Includes all aspects of the training program

Creating an effective training program is a dynamic process that is constantly changing. While it would be nice if the training program creation process would follow an easy, linear progression, the reality is that this process is full of zigs and zags and stops and starts and must take a near infinite number of variables into consideration.

While training program design products and services exist on the market that are more or less plug-and-play, you must realize and understand that if you can enter in just a few fields of information and can get a fully “customized” program, there is no way it can take all of the necessary variables into consideration. As a result, the program is likely lacking in one or more areas. The true value of quality

coaches is that they should always have their eye on the client in respect to the training program and make any and all modifications to the program as needed.

NOTHING IS ‘STANDARD’

Why does one client have two weeks at the same training volume before an increase while another client has three weeks at the same volume before an increase?

Why does one client have a peak training volume of 60 miles/week while another client has a peak volume of only 40 miles/week?

These examples illustrate that when it comes to program design, there is no “standard.” It is important to create a properly constructed training program versus using generic templates. This process takes longer and requires more education on the part of the coach, but it is the only way to ensure that a training program is correct and safe.

PROGRAM CREATION PROCESS

Before you can build out a program for a client, you need to make sure that the event selected to focus on is feasible based on the following factors. Therefore these feasibility steps also represent the first four steps of the program design. The four steps to determine feasibility are:

1. Determine Training Availability
2. Identify Starting Volume (Long Run)
3. Plot Goals
4. Mock Up Long-Run Training Volume

If an event is deemed feasible based on these four steps, continue building out the program with step 5.

STEP 1: TRAINING AVAILABILITY

While steps 1–4 are more or less assessed simultaneously, for the purpose of the certification, they are noted as singular steps.

Before you can start working with clients, you need to know how often they can train and for how long. In order to progress, their training availability must be appropriate for the event type/distance they are training for.

It is rare that a client will have the same availability each week. However, for most clients, there is a general range that they can approximate to train weekly.

If a client's schedule is such that they know exactly which days and times they are available to train, all the better. In this first step, it is a good idea to write down the client's actual time availability and with as much detail as possible. It might look something like this:

Mon	Tues	Wed	Thu	Fri	Sat	Sun
No Availability	30 Min Morning	1 Hour Evening	No Availability	30 Min Lunchtime	1 Hour Morning	30 Min Morning

Figure 14.3 Sample client availability schedule

In regard to misaligned goals, if the client whose time availability listed above has a goal to participate in a marathon, you must explain why the training availability is not sufficient for the goal and recommend that a 10K distance or less would be a better and more feasible option.

It is better to err on the side of caution when selecting an event based on one's training availability. For example, if your client thinks his or her schedule will open up down the road and enable the person to target a half marathon, it is better to go for the sure bet and target a 10K.

RED FLAG!

If a client's goal and training availability are grossly mismatched, such as having three days/week (one hour per day) to train for a marathon, you must inform the person this will not work and select a more applicable goal (i.e., 5–10K).

Realistic Programming

Clients have many obligations other than training, some of which come up unexpectedly. This is the norm, not the exception! It is your responsibility as the coach to develop a program that takes this into consideration, while keeping the client on track to reach his or her goal. For example, if your client has business

meetings that come up often at the end of the day, it is advisable not to schedule critical workouts for that time. You need to work with your client as much as possible to accommodate the person's schedule and personal obligations. However, clients can aim to participate in an event only if they have the time to train for it.

STEP 2: INITIAL TRAINING VOLUME (Long Run)

Once you know that clients' training availability is appropriate for the type/distance event they are training for, you need to know if they are currently training and if so, what their longest run is. For this long run to have value in the creation of the training program, it must be recent, completed without injury, and have been progressively built up to.

Week	Start Date	Long Run Volume (miles)	Training Theme	Event
1	23-Jan	8		

Figure 14.4 Sample initial weekly training volume

Currently Not Running

If a client is deconditioned and/or not currently running, you must be very careful to start the client at a low level and progress the individual slowly to allow the body to adapt from a musculoskeletal standpoint. If a client has never run before and has poor running form, a large focus must be on proper running form/gait. The initial training volume for a client such as this would likely be:

- Run/Walk–20 minutes

A run/walk program has an individual alternate between running and walking at set time intervals (i.e., walk five minutes, run five minutes). As a client progresses through the run/walk program, the run portion becomes longer and the walk portion becomes shorter.

Keep in mind that even if your client is in good aerobic shape, unless the person has been running recently, his or her muscles and connective tissue need time to adapt. This type of client should adhere closely to the preceding volume guideline.

This is also relevant to clients who trained and competed in running events the prior year and stayed in shape during the off-season but have not run recently. Assuming it has been at least one or two months since a client ran, the person should start out at or close to above the initial training guideline. These clients typically adapt faster to running than someone new to the sport, yielding a high rate of progression.

Once a client is up to an hour run/walk program, the individual can advance to a mileage-based (versus time-based) program. The starting run mileage should start out at two miles.

STEP 3: PLOT GOALS

At this point, you have two of the four pieces of information that you'll need to determine if a goal event is feasible. If it is determined that it is not feasible, a client must either select a goal event that is more aligned with training load or not work with you. It is that simple.

This is also the step where you will want to determine which, if any, tune-up races should be added to the program.

Week	Start Date	Long Run Volume (miles)	Training Theme	Event
1	23-Jan	6		
2	30-Jan			
3	6-Feb			
4	13-Feb			
5	20-Feb			
6	27-Feb			
7	5-Mar			
8	12-Mar			
9	19-Mar			
10	26-Mar			
11	2-Apr			
12	9-Apr			
13	16-Apr			
14	23-Apr			
15	30-Apr			
16	7-May			HALF MARATHON (TU)
17	14-May			
18	21-May			
19	28-May			
20	4-Jun			
21	11-Jun			
22	18-Jun			MARATHON (G)

TU = Tune-Up Race

G = Goal Event

Figure 14.5 Goals plotted in training program

Tune-Up Races

Most runners who select a goal race will also select one or more tune-up (preparation) races. These races are important for many reasons: check fitness level, practice fueling, get used to race-day jitters, strategy/pacing, and a lot more. Runners typically do not taper, or do not taper to the same extent for tune-up races as they would for a goal event. As a result, most runners are not completely recovered for tune-up races. Many runners call this lack of a taper *training through a race*.

While a taper is typically not implemented for tune-up races, a runner should still decrease intensity prior to one. The last hard workout prior to a tune-up race should be two to five days in advance (720). The longer the tune-up race, the longer the duration between a hard workout and the tune-up race should be.

Athletes often run races at a casual pace, or at least well below race pace. This is often the case as runners sometimes choose to do a race versus a long run by themselves. It is typically easier for runners to motivate themselves to run a long

distance if they have 5,000 of their closest friends with them. However, it should be noted that when a tune-up race is noted, it means running the race at race pace. A tune-up race is about a lot more than just gaining fitness or getting in a long run.

Trialing the Fueling Strategy

Tune-up races are a great time to practice a fueling strategy. While this can and should be done during training runs as well, everything from prerace, during the race, and post-race can be practiced with all of the dynamics and intricacies of race day! For example, a “nervous” stomach may react differently to the same fuel used in training and on race day.

COURSE SIMULATION

Most races denote the elevation profile on their websites. As such, you can get a pretty good idea of how hard the course is and, specifically, where the most challenging sections of the race are. If an event has a course map but no elevation profile, you can use the website *MapMyRun.com* to create the race map. Clicking on the elevation feature will create the elevation map. This information should not only help shape the overall training process, but the selection of the tune-up race(s) as well.

STEP 4: THE LONGEST RUN

This step is what completes the feasibility aspect of the program. The two elements of this step that are the most important to be aware of are:

- 1. Longest run**
- 2. Suggested long run training volume increases**

LONGEST RUN

The longest run is used to benchmark a client’s preparedness in respect to a goal event. The reasoning behind this is simple – having the ability to run at or

near the full distance of the goal event is a prerequisite to being able to compete and finish a race/event feeling good and with the least chance for injury.

For marathons, peak mileage (long run) is typically attained twice during the program – once during the week before the first taper week and the other three weeks before the first taper week (there is typically a low-volume “long run” in the week between these two peak-mileage long runs).

The typical mileage range for the longest run is:

Half Marathon: 12–13 miles

Marathon: 18–22 miles

10K: 6–10 miles

Why Not Benchmark Weekly Training Volume?

While long runs are a necessity in a training program buildup, training volume (running) may be substituted with other forms of cross-training. As a result, it is possible to perform well at an event with a relatively low overall running volume so long as the long run progression is present in the schedule.

When Do Long Runs Begin?

As noted in the beginning of this module, most training programs assume that an individual is currently running a certain number of miles per week and has also attained a particular mileage for the longest run. Following are typical training program long-run mileage prerequisites in relation to the duration of the training program. This correlates with when the long runs begin.

Event	Current Longest Run	Program Duration
Marathon	6 miles	16–20 weeks*
Half Marathon	4 miles	10–12 weeks*

* The program duration includes the taper weeks

Using these guidelines, if a client is not yet up to the long-run mileage noted, the program will be longer than the noted program duration.

One day per week is typically designated as the “long run” day. Most programs designate the same day during the week for the

long run. However, this is not a requirement. Therefore, when designing a program, any day of the week can be allocated as the “long run” day.

Initially the “long run” may be one mile – however, this will build as the client adapts to the program. Therefore in regard to the question, “When do long runs begin?” the answer is immediately.

SUGGESTED LONG-RUN VOLUME INCREASES

Volume increases in figure 14.6 represent the maximum increases. As such, weeks throughout the long-run build-up may see no increase or an increase smaller than the maximum percent noted on the chart.

Additionally, every three or four weeks the long run should be shortened to one-third to one-half the distance of the prior long-run distance for recovery purposes.

It is likely that as a client progresses through the training program, some of the scheduled training volume increases will be modified based on several factors, including the client’s adaptation to the training and any unforeseen issues such as missed training days.

Current Mileage (Miles)	Maximum Increase
1	1 mile
2	
3	
4	
5	
6	2 miles
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	
21	
22	

Figure 14.6 Maximum individual run mileage increases

If the mock-up shows that a client will not be ready from a volume standpoint, it is advised to select either a shorter event or another event of the same distance but later in the year.

PLOT LONGEST RUNS AND MOCK UP LONG RUN

Following is a mock-up denoting the goal race and tune-up race (half marathon) as well as the two longest runs. Based on this information, a client would have 16 weeks of training before the first of two peak-volume runs.

Week	Start Date	Long Run Volume (miles)	Training Theme	Event
1	23-Jan			
2	30-Jan			
3	6-Feb			
4	13-Feb			
5	20-Feb			
6	27-Feb			
7	5-Mar			
8	12-Mar			
9	19-Mar			
10	26-Mar			
11	2-Apr			
12	9-Apr			
13	16-Apr			
14	23-Apr			
15	30-Apr			
16	7-May			HALF MARATHON (TU)
17	14-May	20		
18	21-May			
19	28-May	22		
20	4-Jun		Taper	
21	11-Jun		Taper	
22	18-Jun		Taper/Race	MARATHON (G)

Figure 14.7 Placement of peak-volume runs

The next step is to plot the starting long-run volume in week one and, using figure 14.6, build up the volume conservatively to see if a client will be ready for the peak-volume run(s). You will have to decide what constitutes a “conservative” volume build-up for your client. Remember, this is a mock-up.

Following are two general guidelines to assist in determining a conservative, mock-up volume progression.

1. There should be a minimum of two weeks at the same long-run distance before increasing distance.
2. Recovery weeks should be spaced every four weeks, with the long-run distance between 30 and 50 percent of the previous week’s long-run volume.
 - a. Ex: Prior week’s long run – 15 miles; Recovery-week long run – 10.5 miles (30 percent)

Week	Start Date	Long Run Volume (miles)	Training Theme	Event
1	23-Jan	8		
2	30-Jan	8		
3	6-Feb	10		
4	13-Feb	7		
5	20-Feb	10		
6	27-Feb	12		
7	5-Mar	12		
8	12-Mar	8		
9	19-Mar	14		
10	26-Mar	14		
11	2-Apr	16		
12	9-Apr	11		
13	16-Apr	16		
14	23-Apr	18		
15	30-Apr	18		
16	7-May	10		HALF MARATHON (TU)
17	14-May	20		
18	21-May	10		
19	28-May	22		
20	4-Jun		Taper	
21	11-Jun		Taper	
22	18-Jun		Taper/Race	MARATHON (G)

Figure 14.8 Placement of long runs

Based on a starting long-run volume of eight miles, the client is properly progressed to meet the peak-volume runs. It is likely that starting at a volume of less than eight miles the client may not be ready for the peak-volume run(s).

At this point, the mock-up has deemed that the proposed marathon goal is attainable and therefore the rest of the training program can be built out. This is noted later, starting with step 5.

Over-Distance Training

This is characterized by training at a longer distance than the race distance. This type of training is most prevalent with 5K and 10K events. Over-distance training should not be part of a marathon program. If the goal event is a half marathon, there is no reason for over-distance training. However, if a half marathon is being used as a tune-up race for a marathon, there will be training days at a greater distance than 13.1 miles. Following are several reasons why over-distance training should be left out of marathon programs:

- **Time:** As it already takes a lot of time to train for a marathon, over-distance training will take time away from other aspects of training, and

more important, it will take time away from other aspects of a client's life (e.g., family time).

- **Injury:** Generally speaking, the longer an individual runs, the greater the chance for an overuse injury. Therefore, even if there is a slight fitness advantage, which is debatable, this advantage would likely be offset by an increased chance for injury.
- **Catch Up:** Training for a marathon is not like a school exam: You cannot cram for it. If the race date is near and a client is unprepared from a fitness or endurance perspective, it is better to pass on the event and choose a shorter event or select a marathon further out.
- **No Need:** Above all else, there is no need to train at a further distance than a marathon.
 - One exception is if your client is training for an ultra-marathon. This particular race distance(s) is not largely focused on in this certification.

STEP 5: INTEGRATING WEEKLY VOLUME/STRUCTURE

In this step, you are determining what the ongoing weekly training volume will be. This takes into account only running miles, not cross-training.

Weekly Volume Percent Increases

Figure 14.9 provides rough guidelines for determining weekly volume percent increases based on the starting volume. These increases represent the maximum percent increases – therefore there often will be just a small increase (or no increase at all) in training volume from week to week.

Weekly volume percent increases are not noted for the purpose of determining if a goal event is feasible, as this is what the prior step is for (Step 4: The Longest Run). Rather, this step is a guide for determining overall weekly volume increases and, more specifically, what the maximum overall weekly volume increases should be. This step is most effectively utilized by determining the maximum overall weekly volume and subtracting the long-run distance. The

remainder of the distance represents the rest of the weekly run distances. The following chart roughly represents the maximum weekly volume increases. Therefore overall weekly volume increases are likely to be less than what is noted in the chart.

Maximum Weekly Volume Increase Chart

CURRENT WEEKLY TRAINING VOLUME (Miles)	MAXIMUM % INCREASE
< 5 miles	50%*
6–10 miles	35%
11–20 miles	25%
21–30 miles	20%
31–40 miles	15%
41–50 miles	10%
> 51 miles	7%

Figure 14.9 Suggested weekly volume increase chart

*If your client is running less than five miles per week, the focus should be on learning proper form and musculoskeletal adaptation.

Identify Training Themes – (optional)

In conjunction with adding weekly volume to a program, training themes can be added as well. This is not a mandatory step but it can greatly assist in identifying areas of focus throughout a training program.

Add Recovery Weeks

Minimally, every three weeks should be followed by a recovery week.

- Characterized by a 30–50 percent reduction from the previous week's volume/distance.
- Intensity during recovery periods should be no higher than an RPE of 6.
- Recovery weeks should coincide with long-run recovery weeks.

Determine Weekly Cycle

This directly coincides with the placement of recovery weeks as this step determines how many training weeks are in between recovery weeks. While there are no rules per se, it is advised that three-week cycles be used if a client is

overtrained, coming back from illness, or prone to burnout or injury. Four-week cycles are the most common while five-week cycles are typically reserved for clients who have a relatively low training volume with lots of rest built into the program or an elite runner who has proved able to handle a high workload. Figure 14.10 is an example of the three different cycles (three-, four-, five-week cycles). The green boxes denote the recovery week.

Keep in mind that a training program does not have to have just one type (i.e., three-week) of cycle. A training program can integrate multiple cycle durations.

3-Week Cycle	4-Week Cycle	5-Week Cycle
22 miles	22 miles	22 miles
22 miles	22 miles	22 miles
15 miles	22 miles	22 miles
	15 miles	22 miles
		15 miles

Figure 14.10 Training cycle durations

Several variables affect the weekly cycle pattern in a training program. The most common are the following:

- Fitness level of client
- Specific point in the training program
- Client's adaptation to training (e.g., not recovering well, prone to injury)

Weekly Volume Pattern Increases

Once the weekly cycles are determined, volume increases within each cycle can be addressed. Generally speaking, it is ill advised to have weekly volume increases for more than two consecutive weeks. As volume increases stress the body, the body needs a time to adapt to the current training volume before increasing it. Constantly increasing the volume week after week likely increases the chance for injury and overtraining.

The following scenarios are examples of how increasing volume patterns could be structured. In figure 14.11, the volume increase is based off the increase percentage noted in figure 14.9.

4-Week Cycle	4-Week Cycle
22 miles	22 miles
22 miles	22 miles
22 miles	26 miles
15 miles	18 miles

Figure 14.11 Training cycle volume distribution

Assessment Checks

Throughout the training process you will want to assess your clients' progress to ensure that they are on target with their goal. This can be done informally using a training log or more formally using set assessments (i.e., MLSS assessment).

Mock-Up Full-Program Volume

Week	Start Date	Long Run Volume (miles)	Weekly Volume (Miles)	Training Theme	Event
1	23-Jan	8	25	Aerobic	
2	30-Jan	8	25	Aerobic	
3	6-Feb	10	25	Aerobic	
4	13-Feb	7	18	Aerobic	
5	20-Feb	10	30	Aerobic	
6	27-Feb	12	30	Aerobic	
7	5-Mar	12	30	Aerobic/LT	
8	12-Mar	8	21	Aerobic/LT	
9	19-Mar	14	35	LT	
10	26-Mar	14	35	LT	
11	2-Apr	16	35	LT	
12	9-Apr	11	24	LT	
13	16-Apr	16	40	LT	
14	23-Apr	18	40	LT	
15	30-Apr	18	40	LT	
16	7-May	10	28	LT/Peak	HALF MARATHON (TU)
17	14-May	20	45	LT/Peak	
18	21-May	10	25	Peak	
19	28-May	22	45	Peak	
20	4-Jun	n/a	31	Taper	
21	11-Jun	n/a	22	Taper	
22	18-Jun	n/a	15	Taper/Race	MARATHON (G)

Figure 14.12 Full-program volume mock-up

STEP 6: DEVELOPMENT OF DAILY/WEEKLY SCHEDULE

Once the structure (duration, cycles, long-run volume, weekly volume, themes) of a program is in place, the weekly training program can be developed. The weekly volume must match that of the previous step. Weekly programs should be designed and sent weekly or biweekly to a client. It is not advised to give your client more than two weeks' worth of programming at a time to allow for client feedback and changes in rate of progression, if necessary.

Following is a weekly template that denotes the following areas:

- Weekly date range
- Training theme (if applicable)
- Morning and evening workouts (if applicable)
- Distance (miles) or time duration of workout
- Pace range
- Total weekly volume

This is an example of what a weekly training program could look like. There may be more or less data depending on what you feel is needed to adequately guide your client through the training process.

	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
TRAINING THEME							
AM Workout							
Time							
Distance							
Pace							
Notes							
PM Workout							
Time							
Distance							
Pace							
Notes							
TOTAL VOLUME							

Figure 14.13 Weekly training program mock-up

INTEGRATE DAILY WORKOUTS

Once the weekly template is established, daily workouts can be integrated.

Integration of Workout Type

Workout types within a training program are repetitious in nature. Therefore, it is suggested you create a database of workouts for your clients. This way, each workout type can be thoroughly explained and the only variables that change are the intensity and volume. How you decide to define them is up to you. However, something simple and intuitive typically works best.

When creating a workout database, categorize exercises by activity. There can be as many workouts under each discipline as needed. For example, if you have six running workouts, they could be R1–R6. By having a workout database, all your client needs to do is look at the referenced workout number to get the workout information and then refer to the weekly schedule for the volume (e.g., time and/or distance) and pace, if applicable.

Creation of a workout database is initially time-consuming, but it is a time saver throughout the training process. A database helps clients, as more specific workout information can be provided for each exercise/workout. If you have an exercise or workout that does not fit into the database, note “custom” on the weekly workout sheet and specify the exercise/workout in an email or over the phone. If you think a workout/exercise might be repeated, add it to the database. Following are examples of how you could denote your workouts for the sport disciplines:

LEGEND	
Y	= Yoga
B	= Bike
R	= Run
RT	= Resistance Train
ST	= Stretch
C	= Custom (not noted in database)

Figure 14.14 Workout legend

Workout Designation

Following is a sample workout. In this example, the following areas are addressed:

- Workout type
- Rate of perceived exertion (RPE)
- Heart rate (HR) zone
 - This can be expressed by other means, such as pace type (e.g., tempo)
- Description/Notes

Run #1 Workout (R1)	
Workout Type:	Long Steady Distance
RPE:	3 or 4
HR Zone:	2
Description/Notes:	This is an endurance-based run designed to focus on your aerobic system and increasing mileage/volume. You should be able to hold a reasonably comfortable conversation while running at this pace.

Figure 14.15 Workout designation

Integration of Daily Workouts into a Weekly Program

Referencing the workout database, integrate workouts into the weekly program.

	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
TRAINING THEME	Aerobic	Aerobic	Aerobic	Aerobic	Aerobic	Aerobic	Aerobic
AM Workout	OFF	R1	R1	C2	R2	R1	C2
Time							
Distance		5 miles	5 miles	20 miles	6 miles	8 miles	25 miles
Pace		9 min/mile	9 min/mile	Easy ride	8:30 tempo	9 min/mile	Easy ride
Notes							
PM Workout	OFF	OFF	ST1	OFF	OFF	ST4	OFF
Time			30 min			45 min	
Distance							
Pace							
Notes							
TOTAL RUN VOLUME	24						

Figure 14.16 Example of integration of daily workouts into weekly program

When integrating daily workouts into the weekly program, you can include as many or as few training guidelines as you want. It is advised to be as detailed as possible so there is little to no interpretation required on the part of a client.

Weekly Distribution of Session Types

How many times per week should the following types of workouts be included in a weekly program?

- Repeats

- Easy runs
- Tempo
- Distance

This question cannot be answered as it is completely based on an individual client and where the person is in the training program. It is advised to select the number of workout types per week based on the following considerations:

- **Client's strengths and weaknesses**
 - e.g., if muscular endurance is a client's weak point, there would likely be an emphasis on increasing mileage intelligently.
- **Injury**
 - e.g., if a client is coming back from a hamstring injury, it is likely that increases in running volume would be minimal and a focus would be on cross-training.
- **Where client is in training program**
 - e.g., is a client in a recovery week or a build-up week? Where a client is in the training program will have a substantial impact on the selection of workout types and volumes.

Break It Down

When determining the number of workout types (e.g., tempo, easy run) per week, it is helpful to begin with the total number of available workouts as well as how much time can be allocated for each specific workout. To determine this, go back to Step 1, *Training Availability*. Using this information, plot out a weekly schedule. Just because a client has a certain number of available days/times to train does not mean that the individual has to train during each available time slot. In fact, this is actually discouraged.

Clients who are training for a running race purely for fitness and/or weight loss often participate in a variety of exercise types such as swimming, cycling, strength training, group fitness classes, etc. Avid runners also partake in these activities to enhance their running. It is encouraged to have your client participate in activities other than running. This helps to reduce the chance for mental burnout and injury and can enhance a runner's performance via cross-training.

STEP 7: PUTTING IT ALL TOGETHER

Some clients (and coaches) want to be able to see the whole program at a glance. While this format typically does not have much in the way of specifics, it can be modified to add more information. This is a fusion of steps 5 and 6.

Below is a mock-up of a hypothetical program based on steps 5 and 6. As the purpose of this layout is to be “at a glance,” instead of denoting workout types as recommended in step 6 (i.e., R4 = Running workout #4), the type of workout and mileage is noted for each day to make it easy to reference.

One caveat to this step is if a client is unable to follow the prescribed workouts for more than a week or more, the rest of the program will likely need to be modified.

Week	Start Date	MON	TUE	WED	THU	FRI	SAT	SUN	WEEKLY RUN MILEAGE
1	23-Jan	5-easy	6-tempo	OFF	6-easy	OFF	8-LSD	OFF	25
	Cross Training		ST		ST				
2	30-Jan	5-easy	6-tempo	OFF	6-easy	OFF	8-LSD	OFF	25
	Cross Training		ST		ST				
3	6-Feb	5-easy	5-tempo	OFF	5-easy	OFF	10-LSD	OFF	25
	Cross Training		ST		ST				
4	13-Feb	4-easy	3-easy	OFF	4-easy	OFF	7-LSD	OFF	18
	Cross Training		ST		ST				
5	20-Feb	6-easy	5-tempo	OFF	6-easy	3-easy	10-LSD	OFF	30
	Cross Training		ST		ST				
6	27-Feb	6-easy	7-tempo	OFF	5-easy	OFF	12-LSD	OFF	30
	Cross Training		ST		ST				
7	5-Mar	6-easy	7-tempo	OFF	5-easy	OFF	12-LSD	OFF	30
	Cross Training		ST		ST				
8	12-Mar	5-easy	4-easy	OFF	4-easy	OFF	8-LSD	OFF	21
	Cross Training		ST		ST				
9	19-Mar	5-repeats	6-easy	OFF	6-tempo	4-easy	14-LSD	OFF	35
	Cross Training		ST		ST				
10	26-Mar	5-repeats	6-easy	OFF	6-tempo	4-easy	14-LSD	OFF	35
	Cross Training		ST		ST				
11	2-Apr	5-fartleks	5-easy	OFF	5-tempo	4-easy	16-LSD	OFF	35
	Cross Training		ST		ST				
12	9-Apr	5-easy	4-easy	OFF	4-easy	OFF	11-LSD	OFF	24
	Cross Training		ST		ST				
13	16-Apr	6-hill repeats	6-easy	OFF	7-tempo	5-easy	16-LSD	OFF	40
	Cross Training		ST		ST				
14	23-Apr	7-Progression	5-easy	OFF	6-tempo	4-easy	18-LSD	OFF	40
	Cross Training		ST		ST				
15	30-Apr	7-tempo	5-easy	OFF	6-tempo	4-easy	18-LSD	OFF	40
	Cross Training								
16	7-May	4-easy	3-easy	OFF	5-tempo	OFF	3-shake out	HALF MARATHON	28
	Cross Training								
17	14-May	8-tempo	5-easy	OFF	7-repeats	5-easy	20-LSD	OFF	45
	Cross Training		ST		ST				
18	21-May	5-easy	5-strides	OFF	5-easy	OFF	10-LSD	OFF	25
	Cross Training		ST		ST				
19	28-May	7-progression	5-easy	OFF	7-tempo	4-easy	22-LSD	OFF	45
	Cross Training								
20	4-Jun	5-easy	6-strides	OFF	6-fartleks	6-easy	8-LSD	OFF	31
	Cross Training								
21	11-Jun	5-easy	4-fartleks	OFF	5-easy	3-strides	5-LSD	OFF	22
	Cross Training								
22	18-Jun	4-easy	OFF	5-strides	3-strides	OFF	3-shake out	MARATHON	41
	Cross Training								

Figure 14.17 Full-program mock-up

- ST=Strength Training

❖ ANNUAL TRAINING PROGRAM

As mentioned previously, most running clients will not be looking for an annual program but rather an event(s)-based program. For clients who are looking for an annual program, it would likely have a similar structure to figure 14.16 – but for an entire year.

❖ SAMPLE PROGRAM SCENARIOS

Clients will come to you with unique situations and goals. You will need to be able to assess each client individually to devise an appropriate and effective training program.

Following are examples of types of clients whom you might come across – all of whom require drastically different programs:

SCENARIO ONE

A deconditioned and overweight client who wants to lose weight and whose motivation is to participate in a 5K at the end of the year.

The first step would be to connect the client with a registered dietitian for assistance in creating a nutritional strategy that will enable the person to lose weight, while still having enough energy to train properly. Depending on height/weight, you might need to be sensitive to the impact of running on the body. This program would more closely resemble an overall fitness/weight-loss routine with a slant toward running-specific training.

SCENARIO TWO

An experienced runner who is trying to qualify for the Boston Marathon.

You need to find out exactly what the individual's strengths and weaknesses are to get the client to the next level. This would involve biomechanical assessments, weekly training program adjustments, nutritional strategies (in conjunction with an RD), and perhaps much more. Working with this type of client will require all of your expertise and, most likely, referrals to other specialists.

If you do not feel comfortable coaching an experienced client, it is best to refer the person to someone else. Just because you are a great coach to one particular level of runner does not necessarily mean you are a great coach to other levels. This is not a weakness, but rather a function of reality; you cannot be everything to everyone. Good coaches know their strengths and weaknesses, and their clientele reflects this.

SCENARIO THREE

Client who works 80 hours a week and has family obligations signed up for a marathon and now needs to find a way to train properly, while not taking time or energy away from work or family.

This type of client is always a challenge to work with because of time constraints, and more than likely, you will not be able to prepare the individual perfectly for the event. It is your job to work with what you have and to prepare the client as best as possible given the circumstances. In this case, it might mean advising the client against doing the goal event and shifting focus to a shorter distance. Important things to ascertain initially are starting fitness level, when the race is, and how much time the person has per week to commit to training. If the date of the event does not align with the amount of training time available, you should recommend a shorter event to focus on.

Clients must be accountable. Just because you prescribe a workout does not necessarily mean a client will follow it. This is not always your client's fault. For example, you might have prescribed an evening training session but maybe a child got sick or a business dinner popped up and the person is unable to do the training. This is life, and you will need to adjust the training program accordingly. However, if these things become habitual or

if a client just doesn't follow the training program, you need to make it clear that the lack of preparation is not your fault but rather due to the person's lack of adherence to the program.

SCENARIO FOUR

A person who is relatively fit wants to do a 10K to stay motivated and to have a goal.

This type of client is quite common. The emphasis of working with this type of client is overall fitness with a slant toward running. Remember, this is not someone whose goal in life is to become an elite runner. The longer the distance of the running race being targeted, the more specialized the focus on running must be.

❖ CLIENT CASE STUDY

SUBJECT: 40-year-old male runner ("Mike")

DATE: April 1, 2015

EXPERIENCE: Has been running for the past four years. In the last two years, he has run a half marathon each year in 2:20 and 2:15, respectively.

GOAL: Run a half marathon in 1:59 or less and lose 10 pounds.

HEALTH HISTORY: Cleared by physician to run and no cardiovascular issues. Clear HHQ and PARQ. No medications. Has had past issues with hamstring tendinosis (left biceps femoris), neck tightness, and groin pain (bilateral).

TIMELINE: Goal event is five months from today (August 1, 2015).

TRAINING AVAILABILITY: During week, one hour in the morning. On weekends, up to three hours on Saturday and Sunday.

CURRENT TRAINING VOLUME: 15 miles/week (six miles longest run) plus two hours of yoga/week.

ASSESSMENT RESULTS:

- Height/Weight: 5'11", 175 lbs
- Body Fat (calipers): 21%
- Perceived Intensity: High pain tolerance
- Resting Heart Rate: 62 bpm
- Maximum Aerobic Heart Rate: 173 bpm
- MLSS (30-minute time trial): 159
- Overhead Squat Screen:
 - Knees moved inward
 - POSSIBLE ISSUE: Weak glute medius (hip abductors), tight hip adductors (could be the cause of groin pain if the adductors are strained)

UESCA CLIENT PROFILE

1. What type of physical activities do you currently participate in (i.e. sports, weight lifting, etc.)? **Running, cycling, yoga**
2. What competitive sports (if any) have you participated in? **Running**
3. Why, when, and how did you get started in running? **4 years ago**
4. Will you be training by yourself or with others? **By myself**
5. What are your short- and long-term running goals? **Run a sub 2-hr half marathon. Lose 10 lbs. Both goals are 5 months from today.**
6. When do you have the most energy, morning or evening? **Morning**
7. Does your schedule vary or is it fairly consistent? **Consistent**
8. What days/times do you have to train? **Week–mornings (1 hr), Weekends–day (up to 3 hrs each day)**
9. How important is structure in your life? **Very important**
10. What aspect of running are you most intimidated by (if any)? **Running too fast and hitting the wall. I have pacing issues.**
11. Name your top three running-related strengths and weaknesses.
 - a. **Strengths: Endurance, high pain tolerance, no orthopedic injuries**
 - b. **Weaknesses: Bad at pacing, eat too much, poor running gait**
12. Do you respond better to tough love, handholding, or somewhere in between? **Somewhere in between**
13. Do you crack, stay the same, or thrive under pressure? **Thrive**
14. What are you better at? Short, hard sprints and intervals or long endurance efforts? **Long endurance efforts**
15. How competitive of a person are you? Rate yourself from 1–10 (10 being the most): **7**
16. How much do you enjoy the “pain” of pushing hard in a race/training? 1–10
9
17. How confident are you in your abilities as a runner? 1–10: **5**
18. You like to be challenged 1–10 (1=untrue, 10=very true): **10**
19. Look forward to hard training sessions 1–10 (1=untrue, 10=very true): **8**
20. Susceptible to mental burnout 1–10 (1=untrue, 10=very true): **3**

SUMMARY/NOTES

Mike has a moderate amount of running experience. As the majority of one's energy while running goes to supporting the body, an emphasis on weight loss will result in Mike becoming a faster runner – in addition to other aspects of training.

His issue with pacing is that he pushes past his limit and often hits the wall. This is compounded by his high tolerance for pain. Therefore an emphasis must be

placed on adhering to pacing guidelines. A GPS watch that has a real-time pace function is recommended so that he can “feel” a particular pace and see how that correlates to the actual pace.

He has very good endurance and low risk for mental burnout and is very disciplined when it comes to workouts, so there will not be much of an issue with program adherence, especially in regard to long runs. However, because he loves long runs, he often avoids short, hard efforts in favor of long, slow runs. To hit his goal time, short, structured, and intense workouts need to be added to his program.

He has a very long gait – especially in front of his body. As a result, his biceps femoris issue may be due to the large eccentric strain placed on the hamstrings to decelerate the lower leg. Shortening this aspect of the stride and possibly converting to a midfoot strike may help.

Since weight loss is a noted goal, referring him to a registered dietician is advised.

Areas of Focus:

- Shortening the stride, possible midfoot conversion
- Refer to RD – weight loss
- Fewer long runs and more speedwork
- Focus on adhering to structured pacing
- Increase range of motion of hip adductors and strengthen hip abductors
- Maintain yoga twice per week

CREATION OF MLSS ZONES

Based on the steps noted in *Module 5: Training Benchmarks*, the training intensity categories for Mike are as follows:

Recovery: 87–105 bpm

Aerobic Base: 106–123 bpm

Aerobic Development: 124–141 bpm

Threshold: 142–159 bpm

Supra Threshold: 160–166 bpm

Max Aerobic: 167–173 bpm

- These heart rate-based MLSS ranges will likely change as Mike progresses in fitness level throughout the training program.

OVERALL PROGRAM STRUCTURE

Taking into consideration the date of the goal race (five months from now) and his current mileage base, Mike has ample time to prepare and lose weight before the race.

Two tune-up races prior to his goal race; one 10K and one half marathon, are integrated into the program.

Based on the starting date and the goal event, Mike has approximately 17 weeks of training. Below is a mock-up of the proposed 17-week training program.

Week	Start Date	MON	TUE	WED	THU	FRI	SAT	SUN	WEEKLY RUN MILEAGE
1	6-Apr	3-easy	3-tempo	OFF	3-easy	OFF	6-LSD	OFF	15
	Cross Training			Yoga		ST			
2	13-Apr	3-easy	4-tempo	OFF	4-easy	OFF	8-LSD	OFF	19
	Cross Training			Yoga		ST			
3	20-Apr	3-easy	4-tempo	OFF	4-easy	OFF	8-LSD	OFF	19
	Cross Training			Yoga		ST			
4	27-Apr	3-easy	3-easy	OFF	3-easy	OFF	4-LSD	OFF	13
	Cross Training			Yoga		ST			
5	4-May	4-easy	4-tempo	OFF	4-easy	3-easy	9-LSD	OFF	24
	Cross Training			Yoga		ST			
6	11-May	4-easy	4-tempo	OFF	4-easy	3-easy	9-LSD	OFF	24
	Cross Training			ST				Yoga	
7	18-May	3-easy	3-easy	OFF	4-easy	OFF	3-shake out	10K	19
	Cross Training								
8	25-May	OFF	4-easy	6-tempo	4-easy	OFF	10-LSD	5-tempo	29
	Cross Training	Yoga				ST			
9	1-Jun	OFF	4-easy	6-tempo	4-easy	OFF	10-LSD	5-tempo	29
	Cross Training	Yoga				ST			
10	8-Jun	4-Easy	4-easy	OFF	6-strides	OFF	6-LSD	OFF	20
	Cross Training			Yoga		ST			
11	15-Jun	4-fartleks	4-easy	OFF	4-tempo	OFF	12-LSD	7-Tempo	31
	Cross Training			Yoga					
12	22-Jun	OFF	4-easy	OFF	4-easy	OFF	3-shake out	HALF MARATHON-TU	24
	Cross Training								
13	29-Jun	4-hill repeats	5-easy	OFF	4-tempo	OFF	14-LSD	4-easy	31
	Cross Training			ST		Yoga			
14	6-Jul	4-hill repeats	5-easy	OFF	4-tempo	OFF	14-LSD	4-easy	31
	Cross Training			ST		Yoga			
15	13-Jul	4-tempo	5-easy	OFF	4-fartleks	OFF	12-LSD	OFF	25
	Cross Training			Yoga					
16	20-Jul	3-easy	3-easy	OFF	4-tempo	OFF	10-LSD	OFF	20
	Cross Training								
17	27-Jul	4-fartleks	OFF	4-strides	OFF	3-shake out	HALF MARATHON	OFF	24
	Cross Training								

Figure 14.18 Case Study Overall Program Structure

**Note: Weekly run mileage includes race distances*

****** Half Marathon-TU = Tune-up race

❖ SUMMARY

- Preformatted training programs should not be used for clients as they lack customization and, therefore, ignore many training variables.
 - Preformatted training programs are a good place to get ideas, but they should not be prescribed to a client without modifications.
- The steps noted in this section relate to the fastest a client could be safely progressed to reach peak weekly volume. Therefore, a slower progression is likely prescribed.
- UESCA recommends a training program that comprises the following three components:
 - Full “At a Glance” program
 - Weekly training programs
 - Workout database
- Many factors affect the degree and frequency of volume increases:
 - Biomechanics
 - Historical training data
 - Injury
 - Current training volume
 - Genetics
- The 10 percent maximum volume increase rule is largely erroneous.
- While maximum individual-session volume increases are noted in this section, smaller volume increases are likely to occur and are based on the individual client.
- Over distance training relates to training at a longer distance than the race distance.
- There are seven main steps to create the training program:
 1. Training Availability
 2. Initial Training Volume (long run)
 3. Plot Goals
 4. The Longest Run
 5. Integrating Weekly Volume and Structure
 6. Development of Daily/Weekly Schedule
 7. Putting It All Together